

Report



SPDC-BFPC UPSTREAM GAS SUPPLY, GATHERING,
EVACUATION & PROCESSING (UGS-GEP) PROJECT

Materials Selection & Corrosion Analysis Gas Gathering & Evacuation Pipelines

Prepared for:

Brass Fertilizer & Petrochemical Company Limited

Doc. No. EA/SPDC-BFPC UGS-GEP/MAT-RPT-001

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SUMMARY PAGE

Client: BRASS FERTILIZER & PETROCHEMICAL COMPANY LIMITED.		
Project: SPDC-BFPC UGS-GEP	Title: Materials Selection & Corrosion Analysis	
Facilities: Gas Gathering & Evacuation Pipelines	Project Ref: EA/SPDC-BFPC UGS-GEP/MAT-RPT-001-A01	

The administration of this document is in respect of the materials specification and integrity management of the Gas Gathering & Evacuation Pipelines aspects of the SPDC-BFPC Upstream Gas Supply, Gathering, Evacuation and Processing (UGS-GEP) Project.

The document defines the basic design concepts, issues, and considerations in the subject matter of materials selection and corrosion analysis, and the attendant specifications and corrosion mitigation systems that will form the basis of the project's Front End and Engineering Design (FEED).

It is also an input & accompaniment to the conceptual study and basic design reports & presentations to DPR as procedural and statutory requirements in the process of securing regulators' approval to progress to the FEED / Detail Design that will lead to the project's FID and implementation.

The report has relied and used data and reference documents provided by client, and if further information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEY WORDS: BASIS FOR DESIGN, PIPELINE, MANIFOLD, PLATFORM, CORROSIVITY, NOMOGRAM

REFERENCES	
Document No.	Description
TO-D-AA-00001	Basis for Design
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Gas Gathering & Surface Facilities Concept Screening Study
EA/SPDC-BFPC UGS-GEP/SVY-RPT-001	Pipeline Route Selection Study
EA/SPDC-BFPC UGS-GEP/PRC-RPT-001	Process Simulation Study – HYSYS, PIPESIM, & OLGA
EA/SPDC-BFPC UGS-GEP/ELE-RPT-001	Manifold Platforms Power Supply Options Study

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Abbreviations

The Following definitions and abbreviations have been used in this document.

Organisational

BB	Bubuwei Bou
BFPC	Brass Fertilizer Petrochemical Company Limited
BYS MEnV	Bayelsa State Ministry of Environment
DPR	Department of Petroleum Resources
EA	Epic Atlantic Limited
FEPA	Federal Environmental Protection Agency
FMEnV	Federal Ministry of Environment
LGA	Local Government Area
NGO	Non Governmental Organisation
SPDC	The Shell Petroleum Development Company of Nigeria Limited

Technical Terms

B/L	Bulkline
BFD	Basis For Design
CAS	Community Affairs, and Security
CPT	Cone Penetrometer Testing, Cone Penetration Test
EDMS	Electronic Data Management System
EIA	Environmental Impact Assessment
EMP	Environmental Management Plan
EPC	Engineering, Procurement, and Construction
FEED	Front End Engineering Design
FID	Final Investment Decision
FDP	Field Development Plan
FP	Full Preparatory

GPS	Global Positioning System
HAZID	Hazard Identification
HRA	Health Risk Assessment
HSE	Health Safety and Environment
IFC	Issued for Construction
IFD	Issued for Design
IFR	Issued for Review
ISO	International Standard Organisation, Isometric Drawing
MTO	Material Take Off
OML	Oil Mining Lease
OPL	Oil Pipeline License
PFD	Process Flow Diagram
P&ID	Piping & Instrumentation Diagram
P.I.	Point of Intersection
PI	Preliminary Investigation

Technical Terms – contd.

POS	Project Overview Statement
PTS	Permit To Survey
QA/QC	Quality Assurance / Quality Control
QC	Quality Control
QCed	Quality Checked
QMS	Quality Management System
RFQ	Request for Quotation
ROW	Right of Way
SPT	Standard Penetrometer Testing
T	Pipeline Tie-in Point
WHIS	Worksite Hazard Information System

Units of Measurements

Ha	Hectare
km	Kilometre
m	Metre
M	Thousand
MM	Million
MMscfd	Million Standard Cubic Feet per Day
MTPD	Metric Tonnes Per Day

1 Administration

This Materials Selection & Corrosion Analysis Study is in respect of the SPDC-BFPC Upstream Gas Supply, Gathering, Evacuation and Processing (UGS-GEP) Project. It stems from the Concept Select Study on the Gas Gathering Concept, and presents the selected design concept as basis for the Basic Design of the project.

It is also an input and accompaniment to the Conceptual Study & Basic Design Memoranda to be presented to DPR as statutory requirements in the process of securing regulators' approval to execute the FEED / Detail Design that will lead to the project's FID and implementation.

1.1 Validity & Change Management

The report has relied and used data and reference documents provided by client. Some of the data are preliminary, based on limited information, but to a large extent, however, are adequate for this phase of work: - conceptual study and basic design.

For example, analogue PVT data was used in the absence of actual PVT, and preliminary pipeline distances were estimated based on desktop route selection study. These design data, and the consequent simulation results will invariably need to be updated during FEED, by which time actual PVT data would have been obtained, and the FP survey done to establish actual distances and elevations of the pipeline routes.

Similarly, the project timelines as set out in the execution schedule was predicated on FID date of June 2018, and that will correspondingly change with actual FID date.

Also, the security situation in the region with its wide spate of vandalism may dictate that certain design provisions be changed during the FEED. The situation will continue to be monitored, and if still perceived to be a cog in the wheel of innovation, some of the ideal designs will be replaced with what will be practically suitable for the situation.

For example, in as much as it is feasible to lay cable from the IPP to the remote manifold platforms to transmit electricity to motorise the platform valves in the Manifold Platforms Power Supply Options, or install solar systems on the platforms, there will be no need to implement such options if the facilities (cables & solar systems) will eventually be vandalised in no time. Instead, it will be rather more appealing to limit the process option to manual valve operations, without the features of valve motorisation that will require electricity.

Epic Atlantic therefore reserves the right to update accordingly, the opinion set out in this report, as further information and / or refinement become available.

1.2 Background and Introduction

The SPDC-BFPC Upstream Gas Supply, Gathering, Evacuation and Processing (UGS-GEP) Project is constituted by Brass Fertilizer & Petrochemical Company Limited (BFPC), in partnership with The Shell Petroleum Development Company of Nigeria (SPDC).

The project scope includes Gas Gathering Pipelines, and a Gas Process Processing Plant that will provide processed gas as feedstock to a Petrochemical Plant and a Power Plant. The Gas Processing Plant will process natural gas gathered from five SPDC fields – Bubouwe Bou, Elepa, Nun River, Seibou, and Opugbene - in OMLs 33, 32, & 36.

The Gas Plant will comprise three trains, (each with a processing capacity of 150 MMscfd) and condensate storage capacity of 50,000 Tonnes. In this respect, BFPC have concluded the Gas Supply & Processing Agreement (GSPA) with SPDC for 270 MMscfd of the natural gas feedstock. The Plants (Gas Plant, Petrochemical Plant, and Power Plant) will all be located in the same complex at Akabel, Brass Island, Bayelsa state of Nigeria, near the shores of Atlantic Ocean.

The Petrochemical Plant will comprise a Methanol Plant of 1.75 MTPA capacity, and a Urea Plant of 2.6 MTPA total capacity, built in two phases of 1.3 MTPA each. It will also have storage capacity for 150,000 Metric Tonnes of Methanol and 100,000 Metric Tonnes of Urea. The facilities complex will also comprise a captive export jetty of 4 berths for up to 35,000 DWT vessels.

The current on-stream date for the project is **December 2020**, and to achieve that, FID is planned for **June, 2018**. In pursuit of these business objectives, BFPC have virtually concluded Project Management & EPC contracts with EIL, Saipem, and TCC, who are internationally renowned contractors in the implementation of the subject projects type.

The Gas Processing Plant, and the associated Gas Gathering System / Pipelines which will transport wellhead gas from the SPDC fields to the Gas Processing Plant, constitute the key upstream enabler facilities for the realisation of the objective supply of processed gas supply as feedstock to the Petrochemical Plant for the product manufacturing of methanol & urea, and to the Power Plant for power generation.

1.3 Scope of Document

The scope of the document covers the basic design concepts, issues, and considerations in the subject matter of materials selection and corrosion analysis, and the attendant specifications and corrosion mitigation systems, structured as follows:

-  Pipelines' Materials Selection & Corrosion Analysis
-  Line Pipe Specification
-  Coating System Selection & Specification, and
-  Cathodic Protection System

1.4 The Pipelines Location

The pipelines location is marshy swampy terrain of Niger Delta of mangrove vegetation, as depicted in the pictorial below.

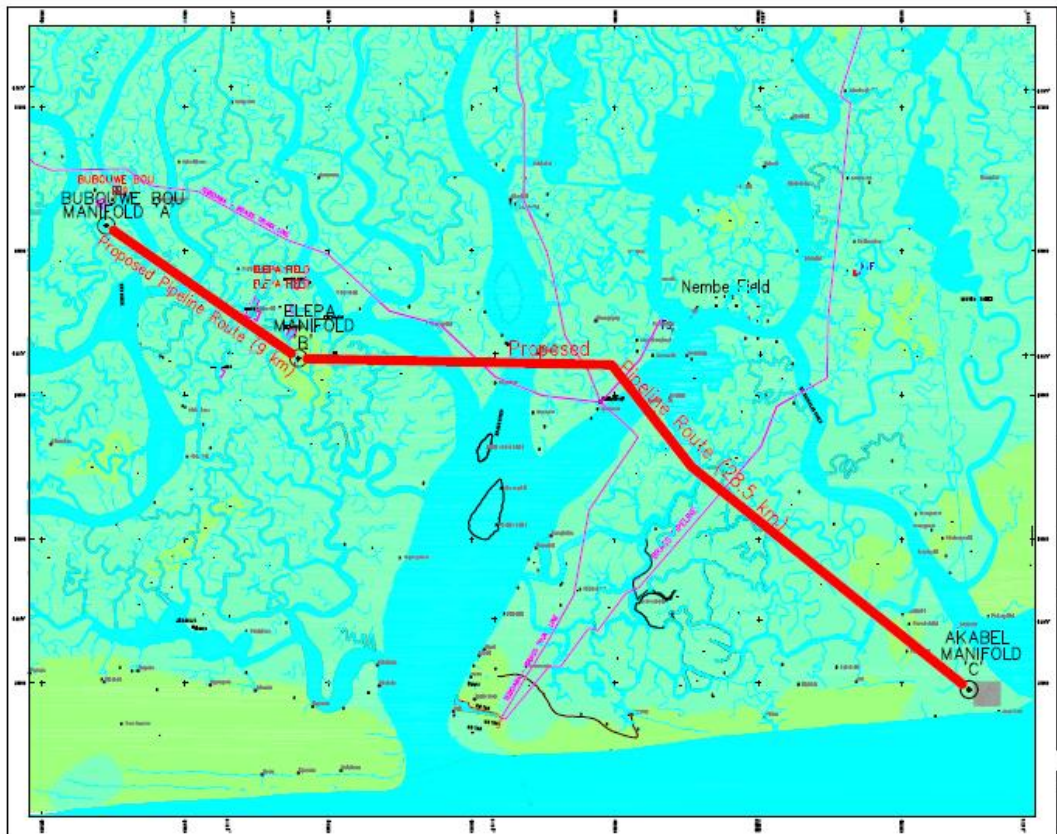


Fig 1.1: Situation Map of the Pipeline Network

2 Pipeline Material Selection And Corrosion Analysis

2.1 Intent & Basis

Intent is Pipeline Integrity, and Materials Selection is the key to this intent.

The bases for the materials selection are the process & environmental conditions, and the operational requirements.

The process conditions constitute the physical conditions, - predominantly pressure and temperature, which the system must contend with in respect of stress; and the chemical conditions emanating from the impurities contained in the natural gas stream, - predominantly carbon dioxide, hydrogen sulfide (*though not in this case*), water vapor, oil, nitrogen, hydrates, which are capable of inducing internal corrosion and erosion of the system.

The environmental condition represents the ambient, which if corrosive will cause external corrosion of the system.

Both corrosions (internal & external) are undesirable and have to be eliminated or mitigated.

The operational requirements constitute consideration of design, construction, operations, maintenance, threats, hazards, risks, safety and economics; and codes and standards that provide mandatory and optional requirements and guidelines.

Similarly, the bases for the corrosion analysis are the susceptibility of the material to corrosion, the corrosivity potential of the transported fluid that will cause internal corrosion to the pipeline system, and the environment which, if corrosive, will cause external corrosion.

These premises will be followed in the materials selection & corrosion analysis process to ensure proper selection of materials and the relational corrosion management system.

2.2 Purpose & Objective

The purpose is three-fold:

1. Identification of best pipeline materials for each of the line segments at optimised life cycle costs, utilising best industry practice.
2. Assessment of the risks associated with the specific selection.
3. Specification of the corrosion management system that will mitigate the risk.

The objective is the optimal specification of materials and the relational corrosion management systems that can contend with the process & environmental conditions, and operational requirements of the pipeline system.

2.3 Materials Selection Principles & Criteria

The material selection process follows an evaluation of the systems materials in consideration with the composition of the streams to which the various pipe segments, pieces of equipment, piping and fittings are exposed as well as the temperatures, pressures and velocities which prevail. The steady state conditions of operation are important but equally any excursions which occur during start-up, shut-down or upsets can be key determinants of materials selection and corrosion management.

The following are factors that have been considered in the selection of materials grade, in order to achieve some degree of protection at optimum costs:

- Design Life
- Properties of Internal Fluid
- Surrounding Environment
- Resistance Level to Corrosion Effect
- Inspection & Corrosion Monitoring Possibilities
- Operating & Maintenance Philosophy, and Degree of System Redundancy
- Mechanical Properties
- Wall Thickness
- Weldability
- Temperature Resistance
- Weight Requirement
- Fatigue and Fracture Resistance
- System Availability Requirement
- Fabrication and Installation Methods and Procedures
- Compatibility with Adjoining Materials
- Market Availability of Materials
- Local Experience (both with the materials and the corrosion management system that will apply)
- Economics

2.4 Materials Options

The following materials are considered pre-qualified for the evaluation.

-  Mild Steel (MS) - Carbon Manganese Steels
-  MS plus Inhibition
-  Various CRA's (Austenitic SS, Ferritic SS, Martensitic SS, Duplex SS)
-  Precipitated Hardened Steel (with Al)
-  Pipeline with Internal CRA Lining or Cladding
-  Composite Materials
-  Non-Ferrous Materials

Despite the corrosion rates which occur when carbon steel is exposed to wet CO₂ it can be a preferred material because of its low cost. In the presence of H₂S corrosion rates may be reduced due to the formation of a protective iron sulphide film. Low temperature carbon steels are also available and are sometimes used to avoid embrittlement when low temperatures might be encountered during blowdown. The corrosion resistance is similar to that of carbon steel.

For the carbon steel pipelines, corrosion allowance and corrosion inhibition form the basis for design life achievement. Where this combination is considered to be insufficient, corrosion resistant alloys are used.

The pragmatic approach therefore rests on the exhaustive investigation of the various material options for the specific function and application.

In this respect, a key factor for the material selection is an up-front activity taking into consideration codes & standards, operational flexibility, cost, availability, sourcing, and the performance of the material for the intended service and application; with due considerations to recommendations of design engineers, materials specialists, manufacturers, installers and operators.

The technical feasibility demonstration for the selection of the appropriate materials follows a four stage evaluation.

1. Feasibility of Mild Carbon Steel, involving the following steps:
 - The prediction of the potential corrosivity of the bare carbon steel
 - Calculation of the Corrosion Allowance (CA) Life cycle period
 - Determination of the acceptance of the selected material (on a **Green**, **Yellow**, and **Red** traffic light colouration system).
2. Feasibility of Mild Carbon Steel, through the following steps:
 - The prediction of the corrosivity

- Calculation of the improved or new corrosivity using the 85% inhibition efficiency and 10% non – availability.
 - Determination of acceptance of the materials on traffic light colouration
3. Feasibility of CRA materials through the following steps:
- Lifetime Cost Cycle (LCC) comparison of various CRA materials with MS
 - Selection of the CRA on the basis of NPV economic evaluation
 - Acceptance determined on traffic light colouration.
4. Optional Process for Modification of the Fluid Content through the following steps:
- Dehydration of the Wet Gas, at the Manifolds or
 - The Removal of CO₂ from the Wet Gas at the Manifolds
 - Brief Review of the processes and systems available
 - Define critical parameter limits to achieve corrosion control goals
 - Select one of the options on the basis of economic review
 - Define the parameters for design

The report will be run stepwise with the materials' acceptance defined in traffic colours.

Red represents: rejection

Yellow represents: not totally acceptable because of some few limitations including costs, properties, maintainability etc.

Green represents total acceptance of the material

2.5 The System & Fluid Under Consideration

The system under consideration is the pipeline network as shown in fig 2.1 below.

It comprises the following pipeline segments, coming into service in phases, and transporting raw wellhead gas to the Gas Processing Plant:

- 24" x 28.5 km ELEP – AKAB Trunkline
- 16" x 9 km BBOU – ELEP Bulkline
- 24" x 45 km NUNR – ELEP Bulkline
- 20" x 20 km SEIB – NUNR Bulkline, and
- 24" x 37 km OPUG – BBOU Bulkline
- Each of the five Field Manifold Headers of about 10 m length.

The fluid transported is high pressure category D/E fluid (flammable fluid), in the pressure & temperature regions of 100 barg, and 60 °C, and containing CO₂ that causes corrosion.

However, it contains no H₂S that causes sour corrosion, or Hg that causes liquid metal embrittlement (LME) of alloys, or Naturally Occurring Radioactive Materials (NORM) which can accumulate to render the system radioactive over time.

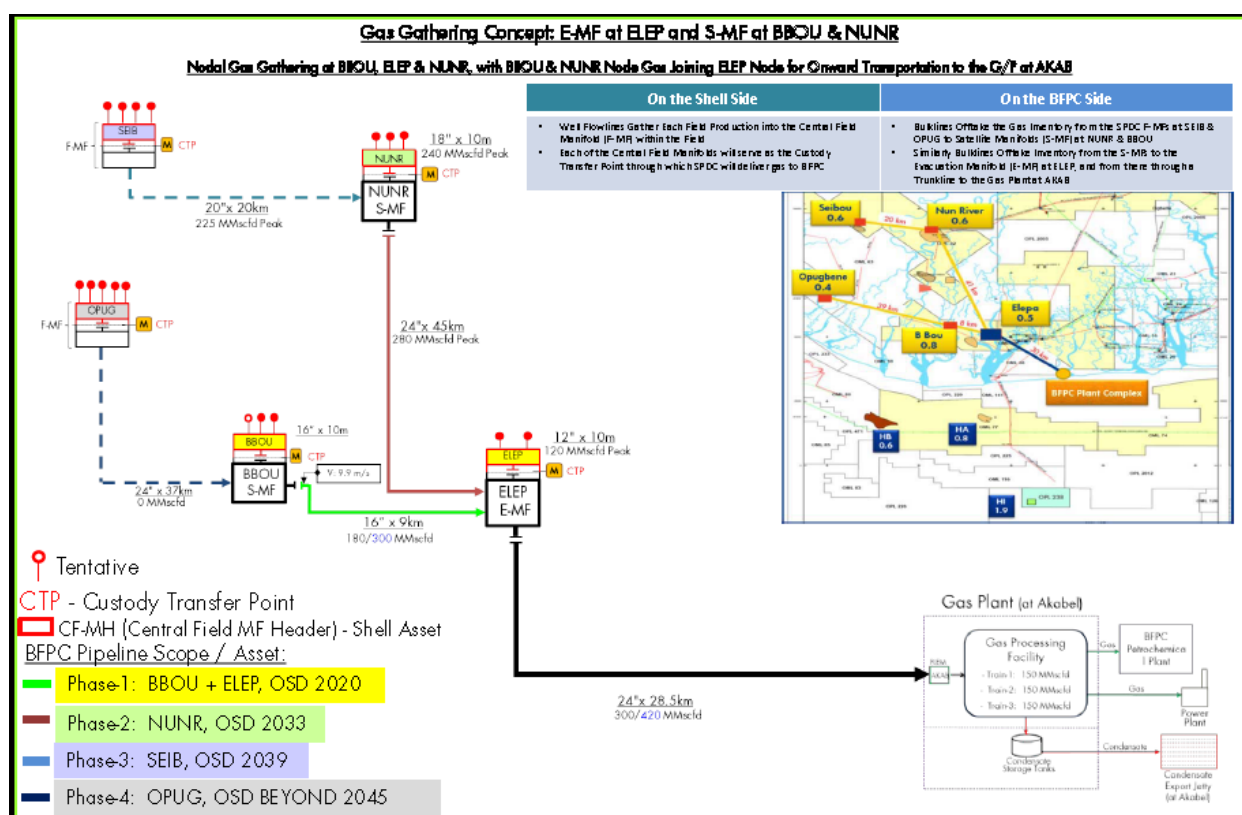


Figure 2.1: The Pipeline Network

2.5.1 The Fluid Impurities Composition

The impurities composition in the various reservoir fluids are shown in the table below:

Table 2.1: Reservoirs Conditions & Impurities Compositions (Water Saturated Compositions)

Component	Impurities Compositions - Mole %				
	BBOU	ELEP	NUNR	SIEB	OPUG
N ₂	0.07	0.00	0.00	0.00	0.10
CO ₂	0.59	2.17	0.77	1.53	2.21
H ₂ S	0.00	0.00	0.00	0.00	0.00
H ₂ O	0.34	2.19	0.63	0.95	1.08
Total	1.00	4.36	1.40	2.48	4.11
Reservoir Conditions					
Reservoir Pressure (bar)	223.8	476.6	500.6	450.6	356.9
Reservoir Temp (°C)	78	123	102	117	113.8
pCO ₂ (bar)	1.32	10.34	3.85	6.89	7.89

At the deducible partial pressures of CO₂ (pCO₂) above, the gas streams will result in significant corrosion of the pipeline system, judging from both the corrosion prediction calculation results from DeWaard-Milliams Model, and its CO₂ Corrosion Nomogram shown below.

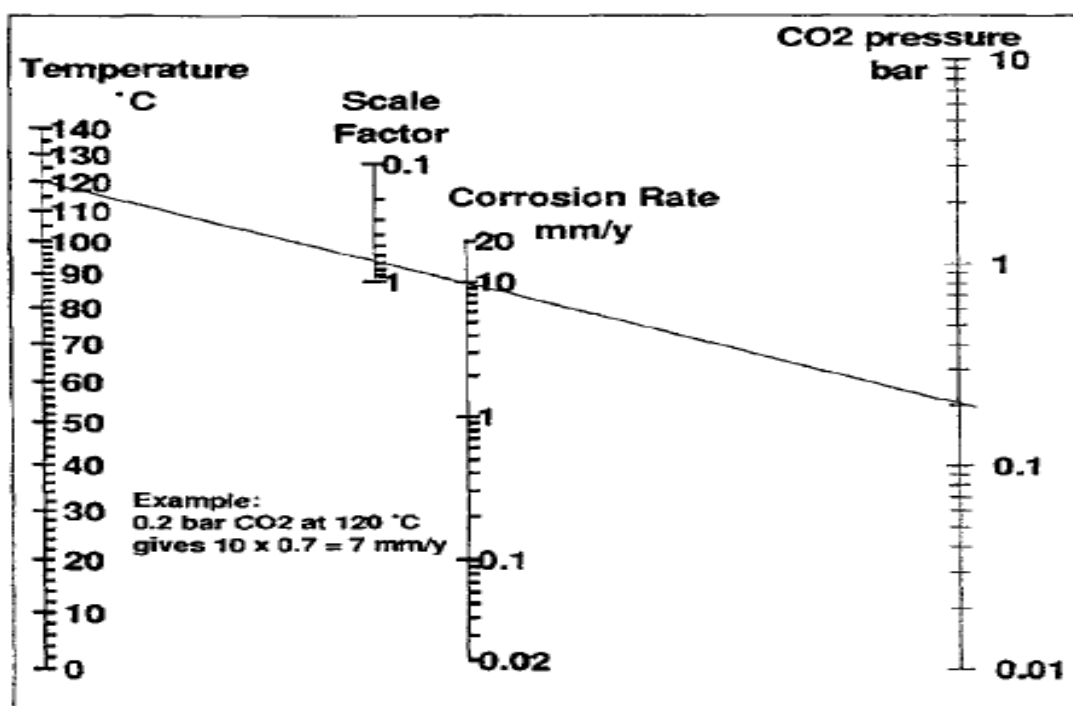


Fig 7.2: DeWaard-Milliams CO₂ Corrosion Nomogram

2.5.2 Issues & Considerations

Of the essence will be pressure containment by the system as well as corrosion prevention / mitigation in the system; and in that respect the following issues were given due considerations:

1. Materials of good strength & properties to withstand the high pressure under consideration and corrosive nature of the fluid to be transported.
2. High Pressure Transportation of Category D/E Fluid (flammable fluid), in the pressure region of 100 barg, and temperature region of 60 °C.
3. The Gas is potentially corrosive on account of its CO₂ content; and water, NGL, & particulates that are inherently present
4. CO₂ content is up to 2%, which for the empirically established 0.1 bar partial pressure threshold for significant CO₂ corrosion, implies corrosion of unprotected mild CS at pressures above 5 barg
5. The CO₂ content of up to 2%, renders the gas corrosive at pressures higher than 5 barg (corresponding to the empirically established 0.1 bar partial pressure threshold for significant CO₂ corrosion), - prompting the need for CRA materials or matching corrosion management design.
6. The Carbon Steel option would mean more internal wall thickness plus corrosion allowance
7. Internal corrosion will be substantial for which the traditional (normal) corrosion allowance of 3 mm will be insufficient provision.
8. Higher corrosion allowance, up to the traditional maximum of up to 10 mm will therefore be considered, and in that respect, high yield strength material with smaller wall thickness demand will be used, so as to keep the overall design wall thickness as small as possible.
9. Ambient environment is corrosive owing to joint contribution of humidity, acidity, and oxygen in air, and known potentially corrosive nature of the soil in the area.
10. Temporal Protection of the linepipes / pipelines, - that will give progressive protection during the period of construction, until the ICCP system is put in place

2.5.3 Corrosion Rates in the Wet Gas Pipelines

The methodology that will be used for the selection of the material will be on the basis of internal corrosion impact on the pipeline and will be as follows:

- A. The Calculations for Corrosion Allowance and Comparison with CRA's through:
 - ❖ Potential Corrosion Prediction of MS Material utilising:
 - De-Waard and Milliams model (NACE 91 model) for MS Corrosion Rate Calculations and /or
 - De- Waard – Milliams Nomogram checks and further comparison of rates with adjustment for gas rates incorporated
 - ❖ Calculations of the Corrosion Allowance for the MS and determination of the risks (failure /replacement Likelihood)
 - ❖ Application of inhibitor, considering 85% Inhibitor Efficiency and 10% non-availability in the corrosion rates calculations (based on local experience and NACE standards), and further determine the improved risks
 - ❖ Comparisons with alternative materials in terms of Life Cycle Costs {LCC}
 - ❖ Comparison with selected CRA's in terms of properties.
- B. Hydraulic Studies for the Pipeline Diameter (Normally, carried out Separately)
- C. B31.8 Wall Thickness Calculations.

The wall thickness is normally determined by installation stresses and service pressures, the design factors, the CA, and the fabrication tolerance.

2.5.4 Further Considerations of Additional Processes Selection by Fluid Modification

The transported fluid may be treated to reduce moisture & carbon dioxide contents which are the main causes of corrosion for the system under consideration, - through the following ways:

- ✚ Use of Glycol / Desiccants to reduce moisture below the threshold that supports corrosion
- ✚ Use of Membrane Separation, Physical or Chemical Solvent to Removal of CO₂

However these processes will mean installation of facilities on the wellhead platforms, which will be very challenging in terms of cost, environmental disposal in terms of waste water into the inlet water ways and the big foot print in the manifolds with increased security and intervention needs; and are therefore not considered viable in this situation.

2.6 Information and Data

Various information are needed for the compilation and analysis to attain the internal corrosion rate, compare options and select the most suitable material for each of the pipelines. Information and their sources are given thereof:

2.6.1 Saturated Gas Compositions

Table 2.2: Saturation Gas Compositions

Composition	Mole Weight	Reservoir Composition (Mol %)				
		BBOU	ELEP	NUNR	SEIB	OPUG
N ₂	28.0	0.07	0.00	0.00	0.07	0.10
CO ₂	44.0	0.59	2.17	0.77	1.53	2.21
H ₂ S	34.1	0.00	0.00	0.00	0.00	0.00
C ₁	16.0	92.61	79.87	72.37	77.84	67.24
C ₂	30.1	3.15	6.00	7.63	5.87	7.88
C ₃	44.1	1.01	2.53	4.82	3.37	4.76
iC ₄	58.1	0.61	0.73	1.45	1.21	1.94
nC ₄	58.1	0.44	0.79	1.96	1.37	1.06
iC ₅	72.2	0.25	0.44	0.94	0.82	1.15
nC ₅	72.2	0.19	0.32	0.76	0.57	0.68
C ₆	86.2	0.23	0.64	1.02	0.91	1.06
C ₇₊	143 - 169	0.53	4.31	7.65	5.48	10.86
H ₂ O	18.01	0.34	2.19	0.63	0.95	1.08

From the table above the main corrosion specie found is CO₂ in Gas i.e. sweet corrosion. The mechanism is CO₂ and oxygen corrosion, - both TOL and BOL attacks are expected

2.6.2 Basic Design Data

The table 2.3 below gives the relevant design data.

Table 2.3: Basic Design Requirement Data for Corrosion

S/N	Description	DATA
1	System	Wet Gas Transportation Pipelines From BBOU, ELEP, NUNR, SEIB, & OPUG Fields
2	Gas Production	300 MMscfd
3	Pressure Ranges	60 -100 bar
4	Velocity	6 – 12.5 m/s
5	Temperatures	44 – 100 Degree C (cf dew points 100°C – 250°C)
6	Mode	Automation
7	Design Life	30 years
8	Design Codes/ Standards	ANSI / ASME 31.8 NACE
9	Pressure Class	ANSI 900 (Class 900) / API 3000
10	Dew Points	100°C – 250°C (cf temps 44°C – 100°C)
11	Water Production	Initial water prod is ca. 420 bpd, but will increase to 10,000 bpd following water breakthrough
12	Condensate Production	ca. 12.5 Mbpd

Important to note is that stream temperature is well above the dew point of gas stream, implying that there will be water drop out that will initiate & propagate corrosion.

2.7 Properties of Optional Materials

The table below gives an overview of the SS CRA materials,

Table 2.4: Properties of CRA Materials

Properties	Austenitic SS	Ferritic SS	Duplex SS	Martensitic SS	Precipitation Hardened
Strength	=	X	XX	XXX	XXX
Wear Resistance	=	=	X	XX	XX
Formability	XXX	XX	XX	=	X
Weldability	XX	=	XX	= =	=
Resistance to General Corrosion	XX	X	XX	=	X
Resistance to Pitting Corrosion	XX	X	XX	=	X
Resistance to Stress Corrosion Chloride	=	XX	XX	=	=

Legend: X - Superior properties = - Lower properties

2.7.1 Pitting Resistant Alloy Number

This number is used to determine susceptibility to pitting corrosion and their critical temperatures.

Table 2.4: Typical PREN, CPT, & CCT

(Source – R.T. Hull, F.A Ramrez, A.I Perez & B.A, Monty (2012))

Material	PREN	CPT (Deg. Centigrade)	CCT (Deg. Centigrade)
316	24 – 26	5 -15	<5
6MO	46	65 -80	30 -60
22Cr Duplex SS	35	20 - 41	17.5 – 25
25 Cr Duplex SS	42	55 -88	33 – 43

Note

PREN – Pitting Resistance Equivalent Number (at 40 and above, no pitting is expected)

CPT – Critical Pitting Corrosion Temperature

CCT – Critical Crevice Corrosion Temperature

2.7.2 Materials Composition

i. API 5L: Composition %

Table 2.5: API 5L: Composition %

P/L Grade	C	Mn	Si	Al x10 ²	Ca x10 ³	Ni	N x10 ²	Cu	V x10 ²	Nb x10 ²	Ti x10 ²	B x10 ³	P x10 ³	S x10 ³
API 5L	0.16	1.56	0.35	4			1.2		7	5		1	3	15

ii. 316 SS :Composition %

Table 2.6: 316 SS : Composition %

Grade	UNS Code	C	Si	Mn	P	S	Cr	Ni	Mo	N
316SSTL	TP	0.08	0.75	2	0.045	0.03	18	14	3	0.10

iii. 6MO SS : Composition %

Table 2.7: 6MO SS : Composition %

Grade	UNS Code	C	Si	Mn	P	S	Cr	Ni	Mo	N
6MO	S31254	0.02	0.70	1	0.30	0.010	20	18	61	0.20

iv. DUPLEX SSTL: Composition %

Table 2.8: DUPLEX SSTL: Composition %

Grade	UNS Code	C	Cr	Ni	M	W	Cu	N
DUPLEX	S31803	0.02	22	5.5	3.0			0.17
	S32205		22.5	5.8	3.2			0.17

v. SUPER DUPLEXSSTL : Composition%

Table 2.9: SUPER DUPLEX SSTL: Composition%

Grade	UNS Code	C	Cr	Ni	M	W	Cu	N
SUPER DUPLEX	S32750	0.02	25	7	4		0.5	0.23
	S32760	0.03	25	7	3.5	0.6	0.5	0.25

2.8 Corrosion Rates Calculations Results

2.8.1 Bare Carbon Steel Corrosion Rates

The corrosion rate prediction calculation were based on De Waards and Milliams model (NACE 91), - given by the formula below:

$$\log V_{cor} = 5.8 - [1710 / (t + 273)] + 0.67 \log (PCO_2)$$

pCO_2 – Partial pressure of CO₂ in the gas in equilibrium with water, bar

t – Temperature °C (the simulated temperatures were given & used

V_{cor} – Corrosion Rate

All rates calculated were cross checked with De Waard and Milliam monogram to ensure accuracy of the rates calculated.

The table below illustrates the summary of the various calculations undertaken

Table 2.10: Summary Of Bare Carbon Steel Corrosion Rates.

CORROSION RATES OF BARE MS IN WET GAS / SOUR SYSTEM											
SN	DESCRIPTION OF LINE	P(CO ₂) in Bar	Design Pressure (in bar)	Design Temp (°C.)	COR RATE (mm/yr)	Recommend d CORR ALL	Life Time Corr. Allowance (30yrs) (in mm)	Recommended CA life time (1n mm)	% LT/Recom.	Comments	Traffic Light
1	20" x 20km SEIB-NUNR B/L	1.64	100	50	4.39	0.1	131.7	3	4390	Very High Corr Rate	
2	18" x 10m NUNR MF	0.77	100	50	2.67	0.1	80.1	3	2667	"	
3	24" x 45km NUNR-ELEP TL	1.095	100	50	3.9	0.1	105.9	3	3530	"	
4	24" x 37km OPUG-BBOU BL		100	50	?	0.1	?	3	?		
5	16" x 9km BBOU - ELEP	0.49	83.35	49	1.15	0.1	34.5	3	1150	"	
6	12" x 10m ELEP MF	1.71	76.12	49	4.43	0.1	132.9	3	4430	"	
7	24" x 28.5 km ELEP - AKAB	0.85	75.96	49.69	2.91	0.1	87.3	3	2910	"	

The results, from the table 2.10 above, indicate that all the MS pipelines reviewed, have very high corrosion rates. The MS bare steel materials are not found to be suitable as the pipeline material for this project. Note that the percentage life time Corrosion Allowance compared to the recommended Corrosion Allowance must be less than 100% for the material to be suitable for purpose. But this is not so. It was found that there were rates comparison between 2667% and 4430%. These are indications of very high corrosion rate. And consequently, this will lead to intolerable high running costs and intervention throughout the project life. Conclusively:

- The bare mild steel is not acceptable in all cases under review.
- Modifiers of the fluid are needed to reduce these very high corrosion rates. Hence, the need inhibition was further reviewed.

2.8.2 Bare Carbon Steel with Inhibition

The table 2.11 shown below is the summary of the corrosion rate prediction for mild steel with inhibition. The inhibition efficiency is considered to be 85%, based on NACE standards. And the availability of the pipeline is 90% (ie for an estimated period of 329 days per year). This is conservative considering that the area is a high flash point for militancy. This may further affect operations.

The formula utilised for calculation is given below

$$CR = F \times CR_m + (1-f) \times C_{ru}$$

F = availability = fraction of time corrosion control system is working (to be obtained from corrosion monitoring) NACE standard 85% to be obtained from monitoring.

CR_m = mitigated corrosion rate.

C_{ru} = unmitigated corrosion rate.

Corrosion rate accuracy generally controlled by accuracy of availability and input data not by model.

CA = corrosion allowance calculation

$$CA = C_{ri} N(1-D/365) + CR_v N D/365$$

Table 2.11: Summary of corrosion Rates of the MS is subject to Inhibition and CA of 3 mm.

Table 2.11

Overall this option showed a lot of improvement on the corrosion rates but still did not achieve the requirement for the design life cycle, except for the 16" x 9 km BBOU – ELEP bulkline.

In conclusion:

- The inhibition has not been able to sufficiently alleviate the corrosion to acceptable levels.
- Only one line is found to be just okay with less stringent standards (16" x 9 km BBOU – ELEP BL)
- Consideration for the need for CRA has now become very relevant on the basis of the corrosion rates obtained.

2.8.3 Corrosion Resistant Alloy (CRA)

The NPV of the CRA is now relevant in comparison with bare carbon steel under inhibition. This is treated in the section 2.10 below.

2.9 Carbon Steel vs Corrosion Resistant Alloys wrt LCC

2.9.1 LCC Analysis For Materials Selection

This LCC is done for further investigation of different materials options at this Engineering stage and to provide the best materials option at the best LCC.

The options selected for this evaluation include:

- Mild Steel Inhibited
- 316 SSTL
- 6MO SSTL
- 22Cr Duplex SSTL
- 25Cr Duplex SSTL
- Titanium

For a balanced comparison and uniform appraisal, 24" lines used are given below with their wall thickness assessed on standard schedule.

Table 2.12: Wall thickness Assessment

PIPE MATERIAL	MATERIAL STANDARD	DIA (INCH)	DIA (mm)	PRESURE class	WALL THICKNESS	SCH
Carbon Steel	A333 Gr6 API 5L x52	24	610	1500	34,93	80
316 SSTL	A312 TP316	24	610	1500	53,34	140
6MO SSTL	A312 S31254	24	610	1500	31,75	80
22Cr Duplex SSTL	A790 S31803	24	610	1500	28,58	60
25Cr Duplex SSTL	A790 S32750/760	24	610	1500	22.23	40
Titanium	B861 Gr2	24	610	1500	22.23	40

Cost Estimate of Options

The following assumptions are made for the costs estimation comparison of the different optional CRA's:

- The line types are standardised at 24" for uniformity
- The length of all lines are 100 m for the LCC calculation
- The expected life used is 25 years
- The inhibited CS corrosion at appendix – indicate the likelihood one or more replacements within 25year period. One replacement at 15th year assumed.
- The uncertainty factor of 2% inflation and discounted rate of 10% for present worth value
- The cost of differed production not included
- Cost of Clamped leaks not included.

Option 1: CAPEX & OPEX Cost Elements for Carbon Steel with Inhibition for 1yr

Table 2.13: Cost Element of Carbon Steel with inhibition

CAPEX Cost Element	\$ meter/weld	Cost for 100 m length(\$)	OPEX Cost Element	\$
Pipe Material	1,756.48	175,648.00	Inhibition	218,846.66
Flanges & Fitting		60,603.12	Inspection	20,000.00
Fabrication	3,378.29	236,459.85	Man Hour	116,482.69
Installation	168.89	16,218.00	Energy	8,736.20
Commissioning	58.24	5,824.00	Spare	15876.60
Design & Admin	58.24	5,824.00		
Total		500,576.97		379,942.15

Option 2: CAPEX & OPEX Cost Elements for 316 SSTL Material for 1yr

Table 2.14: Cost Element of 316 SSTL

CAPEX Cost Element	\$ meter/weld	Cost for 100 m length	OPEX Cost Element	\$
Pipe Material	8,771.20	877,120.00	Inhibition	0
Flanges & Fitting		238,148.84	Inspection	12,736.20
Fabrication	5,411.15	307,397.85	Man Hour	116,482.69
Installation	188.50	18,850.00	Energy	8,736.20
Commissioning	58.24	5,824.00	Spare	41,887.16
Design & Admin	58.24	5,824.00		
Total		1,454,164.69		179,842.25

Option 3 : CAPEX & OPEX cost Elements for 6MOSSTL Material for 1yr

Table 2.15: Cost Element of 6MO SSTL

CAPEX Cost Element	\$ meter/weld	Cost for 100 m length	OPEX Cost Element	\$
Pipe Material	14,472.95	1,447,295.95	Inhibition	0
Flanges & Fitting		909,821.44	Inspection	12,736.20
Fabrication	5,375.65	376,297.33	Man Hour	116,482.69
Installation	188.50	18,850.00	Energy	8,736.20
Commissioning	58.24	5,824.00	Spare	144,729.13
Design & Admin	58.24	5,824.00		
Total		2,763,912.72		282,684.22

Option 4: CAPEX & OPEX Cost Elements for 22 Cr Duplex SSTL Material

Table 2.16: Cost Element of 22 Cr Duplex SSTL

CAPEX Cost Element	\$ meter/weld	Cost for 100 m length	OPEX Cost Element	\$
Pipe Material	9,796.171	97,9617.1	Inhibition	0
Flanges & Fitting		270,239.84	Inspection	12,736.20
Fabrication	5,375.65	376,297.33	Man Hour	116,482.69
Installation	188.5	18,850.00	Energy	8,736.20
Commissioning	58.24	5,824.00	Spare	59,115.00
Design & Admin	58.24	5,824.00		
Total		1,656,594.27		200,070.06

Option 5: CAPEX & OPEX Cost Elements for 25 Cr Duplex SSTL Material

Table 2.17: Cost Element of 25 Cr Duplex SSTL

CAPEX Cost Element	\$ meter/weld	Cost for 100 m length	OPEX Cost Element	\$
Pipe Material	10,134.92	1,013,492	Inhibition	0
Flanges & Fitting		405,359.77	Inspection	12.736.20
Fabrication	5,375.65	376,297.33	Man Hour	116482.69
Installation	188.5	18,8850.00	Energy	8736.20
Commissioning	58.24	5824.00	Spare	84,448,00
Design & Admin	58.24	5824.00		
Total		1,825,647.1		222,403.09

Option 6 : CAPEX & OPEX Cost Elements for Titanium

Table 2.18: Cost Element of Titanium

CAPEX Cost Element	\$ meter/weld	Cost for 100 m length	OPEX Cost Element	\$
Pipe Material	17565.30	1,756,530.00	Inhibition	0
Flanges & Fitting		1520099.12	Inspection	0
Fabrication	5,375.65	364823..0	Man Hour	116482.69
Installation	188.5	188850	Energy	5824.00
Commissioning	58.24	5824.00	Spare	330810.83
Design & Admin	58.24	5824.00		
Total		3,671,985.13		453117.52

2.9.2 Cost Summaries

Note from Appendix – it is predictive and most likely that carbon steel with inhibitor may be replaced at about fifteen years of LCC and at an inflated cost of \$238,137 considering 2% inflation rate for the dollar. Hence one replacement of CS was used for the costing.

2.9.3 CAPEX For the Options

Table 2.19: Summary of Capex Cost estimate

	Carbon Steel with Inhibition	316SSTL	6MOSSTL	22Cr Duplex SS	25 Cr Duplex SS	Titanium
Initial Capex	500,577	1,454,165	2,763,912	1,656,594	1,825,647	3,671,985
15 yrs Later (Repl)	673,711	0	0	0	0	0
Total	1,114,287	1,454,165	2,763,912	1,656,594	1,825,647	3,671,985

Table 2.20: Summary of OPEX

Year	Carbon Steel with Inhibitor	316 SSTL	6MO SS TL	22 Cr Duplex SSTL	25 Cr Duplex SSTL	Titanium
1	784,729	344937	542,188	387,734	426,568	869,079
5	1,931,865	922,563	1,437,340	956,209	1,129,726	2,165,616
10	3,116,738	1,478,059	2,320,535	1522256	1,690.998	3,447,597
15	4,106.320	1,919,870	3,040,325	1,994,844	2,280,713	4,495,262
20	4968,410	2,257,005,	3,668,569	2,389,718	2,769,595	5,412,219
25	5,773,306	2,547,623	4,253,677	2,767,716	3,218,105	6,268.308

Table 2.21: Summary of LCC for Materials

LCC For The Materials Selection (Cumulative Costing) For 100m						
Year	Carbon Steel With Inhibition	316 SSTL	6 MO SSTL	22 Cr Duplex SSTL	25 Cr Duplex SSTL	Titanium
CAPEX Ini	500,577	1,454,165	2,763,912	1,656;594	1,825,647	3,671,985
1	1,285,396	1,799,103	3,306,100	2,044,328	2,252,215	4,541,064
5	2,432,442	2400612	4,201,225	2,612,803	2,955,373	5,837,601
10	3,417,315	2,932,224	5,084,475	3,178,850	3,516,645	7,119,583
15 + Repl. CAP,	5,220,607	3,374,035	5 804,237	3,681,738	4,106,360	8,167,247
20	6;082,681	3,711,170	6,432,481	4,046,612	4,595,243	9,084,204
25	6,887;587	4,001,788	6,917,589	4,424,310	5,143,752	9,940,293

FURTHER REALITY CHECKS ON LCC FOR ELEP – AKAB TL

This LCC comparison is done to cover the following materials

- MS with 3mm CA
- MS with 10mm CA
- Duplex Stainless Steel

And to further ensure that the selected material for the 24"x 28.5 ELEP-AKAB TL, is the best choice.

The cost estimates of materials are based on option 1 phase 1 Epic cost consideration spreadsheet without including the Manifold costs. This plus the construction cost makes up the CAPEX cost.

The OPEX cost estimates are based on Inhibition for one year using CK337 wet gas inhibitor from Nalco, a company in Trans Amadi industrial layout. Other costs elements are Inspection & monitoring, Maintenance, ROW considerations, energy, sparing etc. the following are obtained:

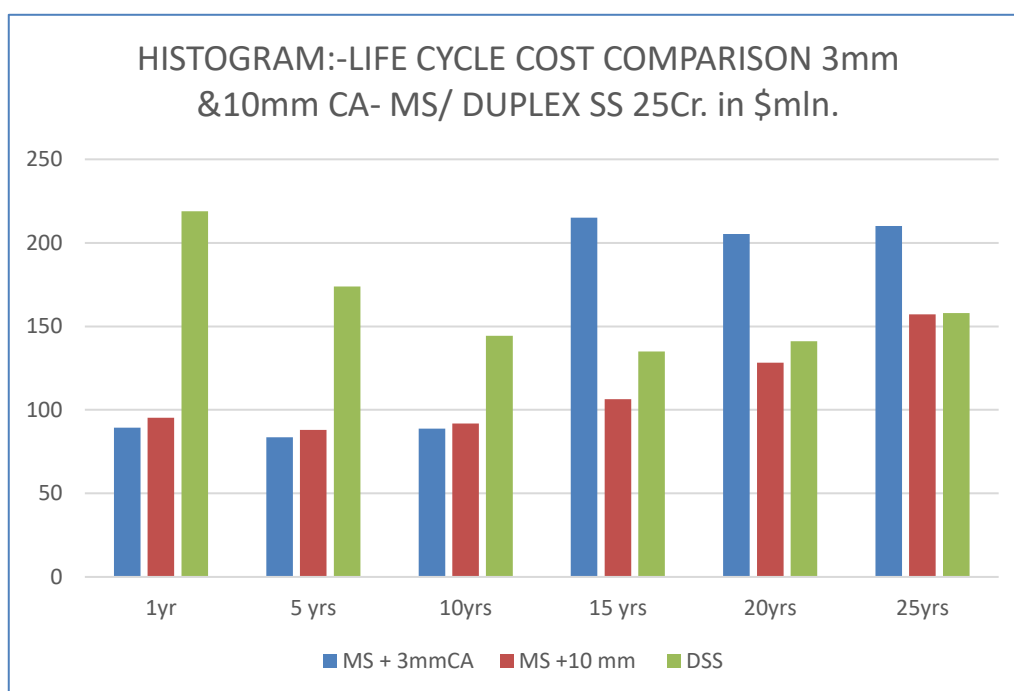
3mm CA cost estimate for CAPEX is \$84.82m and OPEX is \$4.48m per year whilst,

10mm CA cost estimate is \$90.62m and OPEX is \$4.544m

DSS material is estimated as \$214m CAPEX and OPEX of \$4.09m

The NPV was done using dollar inflation rate of 2% and discounted rate of 10% .

The overall outcome is given in the Histogram attached. Hence the material selected is MS with 10mm CA.



Conclusion on the LCC Histogram

Outright the LCC histogram has not favoured the use of Titanium from the year 1 to year 25.

The MS with inhibition is the most favour materials of choice for the first 10 year life of the line. After which period the stainless steels, including the duplex steels, become the materials that merit more consideration. A more comprehensive will cover properties, external environment, community problems and security impact on the costing..

316 SS TL- Austenitic stainless steel has poor properties in surrounding brackish water environment which is saline. It can suffer pitting, crevice corrosion and CSCC (Chloride Stress Corrosion Cracking) in this environment. Again, the risk of hydro testing in aerated brackish water is severe, It is because of conjoint action of stress and chloride corrosion. The area is a flash point for militancy this may attract high cost maintenance that are way above MS steel

This has not favoured its use austenitic stainless steel..

The 22Cr duplex and 25Cr have similar cost profile. The 25Cr is slightly more in LCC but has slightly more superior properties and higher strength. This is what may have makes it the favoured choice for the pipelines .But a big drawback is the initial capital and technical cost outlay as result unforeseen maintenance cost impact due to militancy, as for 316L SS.

Consequently, the short lines within fenced manifolds where security is highly controlled will favour use of the Super Duplex SS. However, for long line that are off – plot only partial acceptance may be considered.. Hence the traffic lights coloration (ref table 2.22) .

Table 2.22: Traffic light Acceptability for CRA's (Duplex SS)

CRA LIFE CYCLE COST (LCC) ASSESSMENT AND ACCEPTIBILITY OF SELECTED MATERIAL							
S/N	PIPELINE	SELECTED MATERIAL	CORR. RATE	P/L LENGHT	INITIAL/CONSTRUCTION ON COST	TOTALITY OF ACCEPTIBILITY IN TRAFFIC LIGHT	REMARKS
1	20" x 20km SEIB-NUNR TL	Super Duplex SS	L	H	M		
2	18" x 10m NUNR MF Line	Super Duplex SS	L	L	L		
3	24" x 45km NUNR - ELEP TL	Super Duplex SS	L	H	H		
4	24" x 45km OPUG-BBOU TL						No info or data for caculation.
5	16" X 9km BBOU-ELEP TL	Super Duplex SS & MS with Inhibition (CA 10mm)	L	M	H		
6	12" x 10m ELEP MF Line	Super Duplex SS	L	L	L		
7	24" x 28.5km ELEP-AKAB TL	Super Duplex SS	L	H	H		

The use 10 mm higher level of CA and Inhibition

This option is analysed for corrosions and the traffic light indications as given below (see table 2.23) The corrosion allowance of 10mm is used. This high corrosion allowance gives much more room for better corrosion management system. It most important, in this case, to select an inhibitor that has a much better efficiency than 85% which was recommended by standards. The selected inhibitor should have an efficiency of 95% and above with a down time of less than 5% per year. The risk to failure to pipelines should be well established, practically, in the first 10 years of the service life. The relevant lines that these measures should be undertaken are:

- 20" x 20 km SEIB – NUNR BL
- 24" x 45 km NUNR – ELEP TL and
- 24" x 28.5 km ELEP – AKAB TL.

It should be emphasised that these lines should be closely monitored and inspected. Any unavailability as result of shutdown shall be accompanied with mothballing. If this is followed and managed strictly we may be able to achieve the goal and acceptability of 10 mm CA.

Table 2.23: Corrosion Rates Of MS Subject to 10 mm CA and Inhibition

CORROSION RATES OF MS WITH INHIBITION											
S/N	Description of Line	P(CO ₂) IN Bars	Design Pressure (Bars)	Temp. (°C)	Corr. Rate (mm/yr)	Recomm. Corr Rate mm/yr	LT CA (mm)	Recomm. CA mm	% LT CA/CA Recomm.	Recommendation	Traffic Light
1	20" x 20km SEIB-NUNR BL	1.64	100	50	1.09	0.1	32.92	3	972	Duplex SS or CO2 Scrubbing	
2	18" x 10 M NUNR MF	0.77	100	50	0.67	0.1	20.1	3	670	Duplex SS	
3	24" x 45km NuNR - ELEP TL	1.095	100	50	0.96	9.1	28.8	3	960	Duplex SS or CO2 Scrubbing	
4	24" x 37 Km OPUG-BBOU BL	?	100	50	?	0.1		3			
5	16" x 9km BBOU - ELEP BL	0.49	83.35	49	0.287	0.1	8.61	3	286	Use MS with inhibition and 10mm ca	
6	12" x 10m ELEP MF LINE	1.71	76.12	49	1.107	0.1	33.2	3	1107	Duplex SS	
7	24" x 28.5km ELEP-AKAB TL	0.85	75.96		0.728	0.1	21.84	3	728	Duplex SS or CO2 Scrubbing	

2.10 Comparison of Selected Line Materials in Traffic Lights

Table 7.2: Traffic Light Assessment

TRAFFIC LIGHTS COMPARISON – based on Cost & Corrosion						
s/n	Line Segment	MS	MS with 3 mm ca plus CI Injection	Duplex SS	MS with 10 mm ca plus CI Injection	Selected Material
1	24" x 28.5 km ELEP – AKAB T/L					MS + 10 mm ca + Corrosion Inhibitor under very strict CMS
2	16" x 9 km BBOU – ELEP B/L					MS + 10 mm ca + Corrosion Inhibitor
3	16" x 10 m BBOU B/L					DUPLEX SS
4	12" x 10 m ELEP B/L					DUPLEX SS
5	24" x 45 km NUNR – ELEP B/L					MS + 10 mm ca + Corrosion Inhibitor under very strict CMS
6	18" x 10 m NUNR B/L					DUPLEX SS
7	20" x 20 km SEIB - NUNR B/L					MS + 10 mm ca + Corrosion Inhibitor under very strict CMS
8	20" x 10 m SIEB B/L					DUPLEX SS
9	24" x 37 km OPUG - BBOU B/L					MS + 10 mm ca + Corrosion Inhibitor under very strict CMS
10	24" x 10 m OPUG B/L					DUPLEX SS

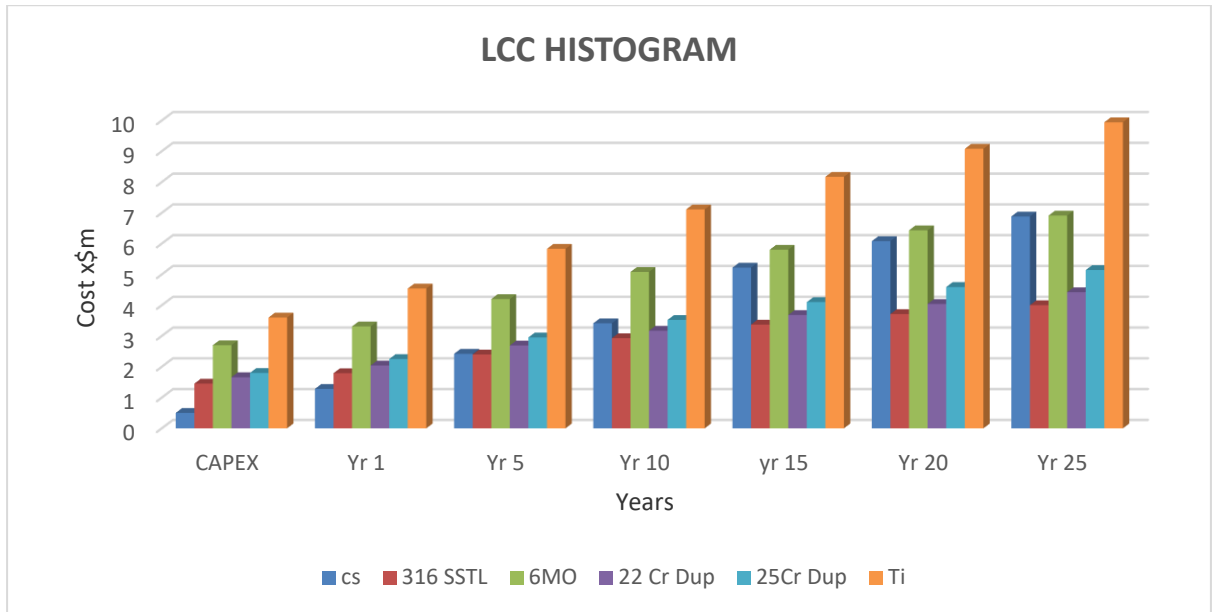
Note:

- The Duplex SS are amber for the long distance pipelines owing to prohibitively high cost of installation, otherwise would be green on purely technical merit.

The cost, properties and, materials selection justifications were built up from bare MS through, MS with corrosion inhibition at 3mm & 10 mm ca (corrosion allowance) to CRA's.

The conclusion that was drawn favoured Duplex SS for the Manifold in-plot short distance lines, and API 5L X65 MS for the long distance pipelines. The API 5L X65 is more acceptable than CRA or Duplex SS, considering cost.

From the corrosivity potential of the transported fluid the MS with 10 mm ca and corrosion inhibitor injection is quite acceptable for the BBOU – ELEP B/L, while the other long distance P/L are only manageably acceptable requiring strict / stringent measures to be taken in terms CMS (Corrosion Management System) in order to achieve the desired service life.



1.1. Materials Specifications

The materials specifications for the various pipeline segments are follows:

- ELEPA – AKABEL 24" x 28.5 km TRUNKLINE / AKABEL MANIFOLD
 - ✓ 24" API 5L X65, wt 31.75 mm for the Trunkline; with the use of Corrosion Inhibitor in the operational phase.
 - ✓ 24" API 5L X65, wt 31.75 mm for the Trunkline; with the use of Corrosion Inhibitor in the operational phase.
 - ✓ API 5L X65 for the pig barrels, and associated kicker line & balance line.
 - ✓ Duplex SS – 25Cr UNS Code S32750 for Manifold In-plot Piping

- ELEPA F-MF – ELEP E-MF 12" x 10 m BULKLINE
 - ✓ Duplex SS – 25Cr UNS Code S32750

- BUBOUWE – ELEPA 16" x 9 km BULKLINE / BUBOUWE MANIFOLD
 - ✓ 16" API 5L X65, wt 31.75 mm for the Bulkline; with the use of Corrosion Inhibitor in the operational phase.
 - ✓ 25Cr UNS Code S32750 Stainless Steel for the short piece 16" x 10 m In-plot Manifold Bulkline that offtakes BBOU production from SPDC BBOU Field Manifold Header for delivery into the BBOU – ELEB Bulkline, either directly or through the Pig Barrel.
 - ✓ Duplex SS – 25Cr UNS Code S32750 for Manifold In-plot Piping

Table 2.24: Traffic Light Comparism

TRAFFIC LIGHTS COMPARISON						
S/N	LINES	MS	MS With Inhibitor + 3mm CA	Duplex SS	MS With Inhibitor + 10mm CA	Conclusion Material Selected
1	20" x 20 Km SEIB-NUNR BL					MS+ inhibition+ 10 mm CA recommended with very strict CMS
2	18" x 10 m NUN R MF					DUPLEX SS
3	20" x 45 Km NUN R – ELEP TL					MS + Inhibition + 10 mm CA recommended with very strict CMS
4	24" x 37 Km OPUG – BBOU BL					NO INFORMATION ON
5	16" x 9 Km BBOU – ELEP					MS +Inhibition and 10 mm CA
6	12" x 10 m ELEP BL					DUPLESS SS
7	24" x 28.5 Km ELEP – AKAB					MS + Inhibition +10 mm CA recommended with very strict CMS

From the above table 2.24 these lines are outright acceptable (green light):

18" x 10 m NUNR MF and 12"x 10 ELEP MF lines will be Duplex SS

Whilst the 16"x 9 km BBOU – ELEP will be Carbon Steel with 10 mm Corrosion Allowance (CA).Three other lines will require a choice between Duplex SS and MS with inhibitor with 10 mm CA. These lines are:

- 20"x 20 km SEIB – NUNR BL
- 24" x 45 km NUNR – ELEP TL and
- 24"x 28.5 km ELEP – AKAB TL.

A preference for MS with Inhibitor and 10 mm CA over Duplex SS is it's lower cost, ease of fabrication, maintainability, and possibly welding issues in brackish, swampy and mangrove environment with high flash point area.

The API 5L X65 is the chosen grade because of the relative thinner wall thickness most of the other PSL1 lines. It also has yield strength and tensile strength advantage with a good carbon equivalent.

2.11 Conclusion and Recommendation.

The cost, properties and, materials selection justifications were built up from bare MS through, MS with inhibition at 3 mm CA (Corrosion Allowance) to CRA's with MS having 10 mm CA with inhibition.

The conclusion that was drawn favoured Duplex SS for the small length lines and for the longer length lines, API 5L X65 were acceptable in comparison to CRA or Duplex SS.

Stringent measures shall be taken in terms CMS (Corrosion Management System) in order to achieve the goals and targets set for three of the MS lines.

The recommended line materials are given on the basis of table 22 above, the hydraulic and thickness calculations were done using API 5L X65 for the Carbon steel. They are as follows:

- 20" x 20 km SEIB – NUNR S-MF BL is 26.19 mm thickness, Sch.80, API 5L X65 with 10 mm CA, no other optional size for this line. The use inhibition applicable in operations phase).
- 18" x 10 m NUNR MF is 10.31mm, Sch. 40, Duplex SS – 25Cr UNS Code S32750.
- 24" x 45 Km NUNR – ELEP TL is 31.98 mm thickness, Sch.80, API 5L X65 with 10 mm CA, with an option is 22" (26.79 mm thick) The use of inhibition applicable in the operations phase.
- 24" x 37 Km OPUG – BBOU BL(For future analysis).
- 16" x 9 Km BBOU – ELEP BL is 22.23 mm, Sch.80, API 5L X65 with 10 mm CA, option is 14" (22.82 mm thick). The use of inhibition also applicable in the operational phase.
- 12" x 10 m ELEP – BL is 9.27 mm thickness, Sch. 40, Duplex SS – 25Cr UNS Code S32750.
- 24" x 28.5 Km ELEP – AKAB TL is 31.98" thickness, Sch. 80, API 5L X65 with 10 mm CA, option is 22" (26.79 mm thick). The use of inhibition is also applicable in the operational phase.

Different operators and national and international standards use different criteria and thresholds to determine whether carbon steel with corrosion allowance or corrosion resistant alloys should be deployed. Differences in corrosion modelling and corrosion inhibitor availability philosophies can lead to carbon steel with minimal corrosion allowance or corrosion resistant alloys being selected for the same field conditions. Even where corrosion resistant alloys are selected factors such as sour service chloride stress cracking exposure to microbes and oxygenated conditions during storage and HISC from cathodic protection can strongly influence the type of CRA selected. Materials selection can also be dominated by mechanical integrity requirements such as strength. This publication examines the various corrosion models and philosophies put forward in different standards and by different operators in the European sector and describes how differing levels of conservatism lead to a wide variation in materials selection. It also examines some of the additional integrity threats

that may occur with the selection of corrosion resistance alloys and the methodologies used to mitigate these integrity threats at the design stage.

In general, the Carbon Manganese Steels (MS) are the first economic choice, due to cost, ease of fabrication and construction as pipeline. It is only when the MS materials are found unacceptable that CRA materials are considered.

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For carbon steel pipelines corrosion allowance and corrosion inhibition form the basis for design life achievement. Where this combination is considered to be insufficient corrosion resistant alloys are used. Different operators and national and international standards use different criteria and thresholds to determine whether carbon steel with corrosion allowance or corrosion resistant alloys should be deployed. Differences in corrosion modelling and corrosion inhibitor availability philosophies can lead to carbon steel with minimal corrosion allowance or corrosion resistant alloys being selected for the same field conditions. Even where corrosion resistant alloys are selected factors such as sour service chloride stress cracking exposure to microbes and oxygenated conditions during storage and HISC from cathodic protection can strongly influence the type of CRA selected. Materials selection can also be dominated by mechanical integrity requirements such as strength. This publication examines the various corrosion models and philosophies put forward in different standards and by different operators in the European sector and describes how differing levels of conservatism lead to a wide variation in materials selection. It also examines some of the additional integrity threats that may occur with the selection of corrosion resistance alloys and the methodologies used to mitigate these integrity threats at the design stage.

In general, the Carbon Manganese Steels (MS) are the first economic choice, due to cost, ease of fabrication and construction as pipeline. It is only when the MS materials are found unacceptable that CRA materials are considered.

The pragmatic approach therefore rests on the exhaustive investigation of the various available material options for the intended function and application. In this stage, a key factor for the material selection is an up-front activity taking into consideration operational flexibility, cost, availability or sourcing and, finally, the performance of the material for the intended service and application.

This report on materials & corrosion is set to address the following:

- Pipeline Materials Selection & Corrosion Analysis
- Line Pipe Specification
- Coating Systems Selection & Specification, and
- Cathodic Protection Systems

It deals with processes and methodology for the selection of appropriate pipeline materials at the optimised Life Cycle Cost (LCC).

For economic reasons, the Carbon Manganese Steels (MS) are the first choice materials for use whenever possible, because of their cheap cost, their ease of fabrication and construction in the pipeline industry. It is only, when the MS materials are found unacceptable after their corrosion analysis that other consideration for CRA's alternatives are given.

The attributes of the project are, for which the following materials are considered as options that will suit or fit:

- ✚ MS
- ✚ MS plus Inhibition
- ✚ Various CRA's (Austenitic SS, Ferritic SS, Martensitic SS, Duplex SS)
- ✚ Precipitated Hardened Steel (with AL)
- ✚ Pipeline with Internal CRA Lining or Cladding
- ✚ Composite Materials
- ✚ Non- Ferrous Materials

Other Materials, such as Composite Materials, Monel (Ni enriched Steel), Non-Ferrous Materials also flavour as alternatives but are not yet tested locally in pipeline construction to present enough justification for their consideration in this selection process. Also some of them come in small sizes and consequently, may involve the use of many more pipelines (say three or more parallel lines) to achieve the same service as the traditional large diameter size, while some (e.g. the thermoplastic composite) have a wide range of failure modes that render them practically unsuitable.

3 Materials Selection And Corrosion Analysis : Coating Materials.

3.1 External Protection of Pipelines

- Coating Material Selection
- Cathodic Protection

3.2 Selection Coverage

- Surface Preparation
- Application of Primer
- Coating Repairs
- Materials
- Storage
- Inspection And Testing

3.3 Background

The proposed BFPC pipelines are to be constructed to supply wet gas from SPDC JV locations widely spread on both, sides of the Nun River at the Brass area, bordered by the Atlantic Ocean at Bayelsa state of Nigeria.

This report forms part of the second section of a three parts report on materials selection for External corrosion Protection which deals with coating materials choice. The concrete addition on the coat is not treated here but separately treated on the main design document because it is considered on the principle of floatation rather than corrosion.

This aspect of the report is a build up from the materials selection report which provide the pipe grade and specifications to be used. The effectiveness of coating is augmented by the CP system for the full pipeline protection.

The coating fabrication on the pipeline is normally carried out in contractor's yard before it is conveyed to site for installation. It should be noted that handling precaution is necessary to eliminate coating damage.

3.4 Scope

The scope of coating materials will cover both MS and duplex SS as indicated below in table 1.

Table 1 Selected

Table 3.1: Pipelines

SN	PIPELINES	SELECTED MATERIALS	THICKNESS AND SPECIFICATIONS
1	20" x 20 km SEIB – NUNR BL	MS + INHIBITION	26.19 mm thickness, Sch.80, API 5L X65 with 10 mm CA
2	18" x 10 m NUNR MF BL	DUPLEX SS	10.31 mm, Sch. 40, Duplex SS – 25Cr UNS Code S32750
3	24" x 45 km NUNR – ELEP TL	MS + INHIBITION	31.98 mm thickness, Sch.80, API 5L X65 with 10 mm CA,
4	24" x 37 km OPUG – BBOU BL		Not Evaluated
5	16" x 9 km BBOU – ELEP BL	MS +INHIBITION	22.23 mm, Sch.80, API 5L X65 with 10 mm CA
6	12" x 10 m ELEP MF BL	DUPLEX SS	9.27 mm thickness, Sch. 40, Duplex SS – 25Cr UNS Code S32750.
7	24" x 28.5 km ELEP – AKAB TL	MS + INHIBITION	31.98" thickness, Sch. 80, API 5L X65 with 10 mm CA

3.4.1 Basis of Design

Risk of external corrosion can be mitigated by applying anti corrosion coating on line pipes to be buried, forming a barrier between the steel and the environment. Application of good coating reduces the C.P current requirement as both {Coating and CP} in conjunction provide external protection

3.4.2 Design Requirements

The selection of coating material is based on

- The selection of materials must meet the accepted standard from an HSE view point
- The material should provide high level of protection for the entire life of the pipeline
- The adhesion between the steel and coating shall be effective throughout the life of the pipeline and the disbondment of the coating minimal
- Durability of the resistance to chemical, physical and biological degradation
- Maximum and minimum temperatures
- Strength and impact resistance
- Ease of repairs of the damage coat
- Compatibility with concrete weight
- Acceptable unit price
- Ease of application
- low permeability to water and salt
- permeability of oxygen low

3.4.3 Design Life

30 YEARS

Design Parameter

Pipelines

Table 3.2: Pipelines

	MATERIALS	PIPE SIZE Inch	PL LENGTH km	MAX OPERATING TEMP. OC	MATERIALS GRADE	COMMENTS
SEIB - NUNR BL	MS	20	20	50-100	Sch.80,API 5L X65	
NUNR MF BL	DUPLEX SS	18	0.01	50-100	Sch. 40, – S32750	
NUNR - ELEP TL	MS	24	45	50-100	Sch.80, API 5L X65 ,	
OPUG - BBOU BL						Not Evaluated
BBOU – ELEP BL	MS	16	9	50-100	Sch.80, API 5L X65	

ELEP MF BL	DUPLEX SS	12	0.01	50-100	Sch. 40, S32750.	
ELEP – AKAB TL	MS	24	28.5	50-100	Sch. 80, API 5L X65	

Coating Materials Information

X - Unaccepted performance

✓ - Accepted performance

Table 3.3: Coating Materials Information

S/N	MATERIALS	OPERATING TEMPERATURES	DISBONDMENT (Life time)	EASE OF APPLICATION	HSE	COMMENTS
1	Asphalt	X	X	✓	X	
2	CTE	X	X	✓	X	
3	Polyethylene Wraps	X	XX	✓ X2	✓	
4	Epoxy/PE/PP	✓ X2	✓ X2	✓	✓	
5	FBE	✓ X2	✓ X2	✓	✓	

3.5 Selected Coat

3.5.1 Coating Material

On the basis of the analysis above, the selection of suitable coating material for this project should be cost effective and also depend on the operating temperature of between 50oc and 100 oc. Hence the higher temperature will favour the use of 3 layer polypropylene in accordance with ISO 21500 -x12011. Besides meeting the technical requirements it is available in the country.

3.5.2 Coating Materials Composition

1st layer epoxy

2nd Layer Adhesive

3rd Layer PP top coat

It should be noted that the qualifications of coatings will comprise:

- Batch Certification
- Coating System Qualifications General
- Procedure Qualification
- Inspection and Test Plan
- Visual Appearance
- Coating Thickness see below table on minimum recommended thickness for coating

Table 3.4: Coating thickness

Design Parameter	Unit					
Pipe size	Inch	10	12	16	20	24
Coating Thickness	mm	2.0	2.9	2.9	2.9	2.9

Other relevant items are:

- Holiday detection
- Impact Resistance
- Indentation Resistance
- Elongation
- Thermal Ageing
- Hot Water Resistance
- C.P disbanding
- Degree of Cure of FBE Primer.

3.5.3 Coating Materials for Bend

Bends are factory coated using 3LPP as it relates to the pipeline

3.5.4 Coating Materials for Field Joint

The recommended field joint is the shrinkage sleeves type. However, the contractor may come with another system. The procedure application must be qualified before use and also proved by BFPC before use.

3.5.5 Coating Materials Repair Works

The Contractor should provide the procedure of the repair methods to be used and also the material for prequalification exercise before approval for use.

3.5.6 Surface Preparation

The line pipe will be prepared to SA2 .5 ISO 8501.4 standards by blasting. It should be noted that for duplex [SS the grit chips](#) should be such that it does not leave any free FE on the surface. [A solution may be the use of Al grit chips.](#)

4 Cathodic Protection Design

4.1 Introduction

Cathodic Protection is one part of a co-joint requirement along with coating application to provide the full external protection of pipelines networks or systems.

This is a preliminary CP design works for the BFPC pipelines that will transport wet Gas from various SPDC locations in the Brass/Nun River areas of Bayelsa state. At this stage only desk top analysis on the protection will be done. And the CP will be temporary until all relevant studies and construction decisions of the pipelines have been made. Chemical, Resistivity Surveys along the ROW, selection of Ground bed locations and types [will be based on general information and assumptions as the true field data and surveys will be conducted before the final design is implemented.](#)

CP design is not done in isolation of the coating applied on the pipelines. The function of the coatings is to reduce the surface area to be protected by CP. The Coating, normally, is intended to form a continuous film of an electrically insulating barrier material over the pipeline "surface to be protected from direct contact with the surrounding electrolytes and to interpose such a high electrical resistance that electro-chemical "corrosion" reactions cannot ready occur. In reality, all coatings, regardless of overall quality, contain holes or holidays that are formed during application or developed during transportation or soil stress and other disbondment from pipe surface including the degradation of the coating during service.

For this CP design the full information on the selected coating material, performance, characteristics, and limitations will be required and considered.

A good understanding of the basic theory of CP is necessary for this work.

Cathodic Protection is a technique to reduce the corrosion rate of a metal surface by making it a cathode of an electrical cell.

The two main ways CP are practically applied are through:

- Impressed Current on the Pipeline
- Sacrificial Anode in Galvanic System with the Pipeline being the Cathode

The following consideration will be reviewed in this design work.

- Criteria for CP
- Selection of Type, Size, and Spacing of the System
- The Needed Information that will Influence the Choice of the System (sacrificial or Impressed Current)
- Total Current Requirements
- Special Conditions requirements

A more comprehensive requirement will be the following;

- Corrosivity of the Environment and Soil Structure and resistivity Survey
- Coating Protection level/ Anticipated Deterioration level
- The metal / Alloy used For the Pipeline
- The Size of the Pipeline and its Ability to Conduct CP current
- The Presence of Foreign Structures , Crossing etc
- The Presence of Stray Current Protection
- Chemical Reaction Survey and
- Attenuation Survey.

4.2 Project Description

It is envisaged that the Brass fertilizer plant (owned by BFPC) will come into production in 2018 and later on to full capacity production of:

- 4,400 MTPD of Ammonia
- 3000 MTPD of Methanol
- 7,700 MTPD of Urea
- Nitrite Fertilizer Product

The plant and its infrastructural facilities will be located bordering at the Atlantic Ocean in Brass Island, Bayelsa state of Nigerian. It will have facilities widely spread on both sides of the River Niger (West and East). The plant feedstock will be Gas from SPDC JV wells at BBD3 Analogue , Nun River, Seibou and Tollogbene fields.

The wet gas will be transported from SPDC's manifolds through pipelines at these identified locations through the bulklines via the main pipelines to the Gas plant via the main Gas Transmission lines to the gas plant where it is treated , dried and pass to the Fertilizer plant as shown in the Fig 4.1



Fig 4.1: Propose Pipelines' Routes

4.3 Project Objective

The objective of this project is to carry out Cathodic Protection preliminary design of the BFPC wet gas bulklines and pipelines from SPDC to the BFPC Fertilizer Plant

4.4 Purpose

The document details the preliminary cathodic protection for the external protection of corrosion of the BFPC bulklines and pipelines.

4.5 Scope

The lines on Table 4.1 below are the scope of work

Table 4.1: Pipelines Information

SN	PIPELINES	SELECTED MATERIALS	COMMENTS
1	20" x 20 km SEIB – NUNR BL	MS	10mm CA
2	18" x 10 m NUNR MF BL	DUPLEX SS	No CA
3	24" x 45 km NUNR – ELEP TL	MS	10mm CA
4	24" x 37 km OPUG – BBOU BL		Not Evaluated
5	16" x 9 km BBOU – ELEP BL	MS	10mm CA
6	12" x 10 m ELEP MF BL	DUPLEX SS	No CA
7	24" x 28.5 km ELEP – AKAB TL	MS	10mmCA

5 Cathodic Protection Design (Definition)

5.1 General Definition

The **Contractor** is the party which conduct all or part of the design. The Principal may undertake all or part of the duties.

The **Manufacturer/supplier** is the party which manufacture or supplies equipment and services to perform the duties specified by the contractor.

The **Principal/Company** is the party which contracts the project and completely pay for it' design and construction. The Principal will generally specify the technical requirement. The Principal may also include an agent or consultant authorized to act for and on behalf of the Principal.

The word **shall** indicates requirement.

The word **should** indicates recommendation.

5.2 Abbreviation

API	American Petroleum Institute
ASTM	American Society for Testing and Material
CP	Cathodic Protection
DEP	Shell Design Practice
GB	Ground Bed
ICCP	Impressed Current Cathodic Protection
IJ	Isolation Joint
ISO	International Standard Organisation
NACE	National Association of Corrosion Engineer
PMT	Project Management Team
SPDC	Shell Petroleum Development Company

6 Cathodic Protection Design (Technical Information/Data)

6.1 Design Parameters

a) Pipeline Information

Table 6.1: Pipeline Data

S/N	PIPELINE	LINE DIA.	LINE LENGHT	DESIGN PRESS	AMB. TEMP.°C	TEMP. °C	THICKNESS + CA (mm)	SELECTED COATING TYPE
1	SEIB – NUNR BL MS	20"	20 km	3,000	35	50-100	16.19 +10= 26.19	3LPP
2	NUNR MF BL (DUP. S/STEEL)	18"	10 m	3,000	35	50-100	10.31mm	3LPP
3	NUNR – ELEP TL MS	24"	45 km	3,000	35	50-100	21.98 +10 =31.98 or 16.79 +10 =26.79	3LPP
4	OPUG – BBOU BL	24"	37 km					
5	BBOU – ELEP BL MS	16"	9 km	3.000	35	50-100	xxxxx	3LPP
6	ELEP MF BL (DUP. S/STEEL)	12"	10 m	3.000	35	50-100	9.27	3LPP
7	ELEP – AKAB TL MS	24"	28.5 km	3,000	35	50-100	21.98 +10 =31.98 or 16.79 +10 =26.79	3LPP

b) Design Life - 30 years

c) Expected Coating Disbondment in 30 years

Polypropylene coating breakdown value is 5 to 7% at 25 years the mean for the period is 2% from DnV RP B-401 standards with expected 0.75 to 1.5% per year thereafter.

- d) Reference Electrode Potential Ranges for Steel Protection (starting with the criteria for Protection to the beginning of overprotection)

Table 6.2: Criteria for Protection Vs Reference Electrode

Environment	Reference Electrodes		
	Cu/CuSO ₄ (mV)	Ag/AgCl (mV)	Zinc (mV)
Free Seawater Environment	-850 to -1000	-800 to -1050	+250 to +0
Anaerobic Environment	-950 to -1100	-900 to -1050	+150 to +0
High strength Steel UTS>700N/mm ²	-850 to -1000	-800 to -900	+250 to +100
Land/ Swamp Environment	-850 to -1150	Not used for land	Not used for land

e) **Environmental Condition**

Environment – mangrove swamp and brackish water of Resistivity 1000 -5000 ohm – cm (assumed)

Mean Minimum temperature	230C
Minimum temp	180C
Mean Maximum Temp	310C
Ground Temp	25 -27.50C
Black Body Temp	800C
Design Ambient Temp-Max	430C
Design Ambient Temp Min	180C
Humidity	100%
Site Altitude	Sea level
Lightning	Frequent in the Raining Season
Average Annual Rainfall	3800 in

Mean Maximum hourly Rainfall 100 in
Design Wind Speed 35.6 m/s
Corrosion Environment Brackish Swamp/Waters (50 -100 ohm-cm)

f) Environment / Resistivities Vs Corrosion Current per M^2

The assumption made for the current density for bare steel metal areas for this desk top calculations is 22mA/m² at the assumed resistivities of 1000 – 5000 ohm-cm of the swampy, brackish water environment closed to the brass sea

g) Coat efficiency, Mean Coating Breakdown and coating Breakdown at end of life time

Table 6.3: Coating Breakdown for Design Purpose

SN	Type of Coat	Efficiency on Yard Fab. (%) (Initial)	Breakdown after 30 years (%)	Mean Coating Breakdown (%)
1	Epoxy 3LPE or 3LPP	99.9 - 99.5	7- 10	2
2	Reinforced Coal Tar or Asphalt Enamel	98	10 -20	5-10
3	Polyken Wraps	95	50 (15yrs)	15 -30
4				

7 Cathodic Protection Design (Basis For Design)

7.1 Basis For Design

- The external protection of the pipelines along with the coat
- Preliminary cathodic protect shall be provided during the construction period of the pipelines
- The final CP system shall be implemented following the required field surveys for chemicals, soil resistivity for the ROW and brackish water crossing.
- Resistivity survey shall also be done for the selection of Ground bed locations.
- Design life is 30 years
- Design life for temporary CP system during construction is 3 years
- Choice of CP powering will depend on availability of power in the manifold
- All pipelines will be buried or submerged
- Soil resistivity for the temporary CP is assumed to be 1,500 Ohm-cm
- Resistivity of carbonaceous backfill is assumed to be 0.5 ohm-m
- The pipelines will be coated with 3LPP and in good quality
- The temporary CP current estimates will be made on deck-up choice
- Reference electrode is saturated Cu/CuSO₄ for soil and Ag/AgCl for seawater

7.2 Choice Of CP System

The CP system shall be used for the external protection of pipelines against corrosion. In this specific case there is no margin for error in term of corrosion based on the limitation on the selected material corrosion tolerance (No room for external corrosion). A three year temporary CP shall be incorporated during the construction. This design will be concluded in this document.

The factors that will influence the choice of the CP system will include:

- Availability of Electrical Power System
- Pipelines will be buried or Submerged
- Pipeline / Environmental Considerations
- Polarisation Limits

The choice that are available are:

- The Type of CP System – ICCP, SA, or Hybrid
- Electrical Power System – National Electricity Board, Company Supplied Electricity, Remotely Powered Generators (Solar Powered, Thermo- Electrical, Wind Powered, Batteries ,Turbo Generator, Fuel Cells etc)
- Ground bed locations Selection,
- Anodes Choice

7.3 Current Requirement

7.3.1 Current Density

Current density is the corrosion current released per square area of a material under electro-chemical influence. It follows that the total corrosion current will be the current density multiply entire surface area. This area can be significantly reduced by application of coat. Allowing only for exposed areas that the coat has not relevantly affected to corrosion. Cathodic Protection is the application of the reverse cathode current to reduce or arrest the electro chemical corrosion reactions.

Hence, the determination of the total current is used to evaluate the design properties using the given equation for pipeline:

$$I_c = JF2\pi rL \text{ —————eq. 1}$$

Where J – is the design electrical current density

F – Coating Breakdown Factor

$2\pi rL$ – Surface Area

7.3.2 Current Demand Calculations

From above equation 1, the design electric current demand for each steel bulklines & pipelines taking into cognisance the coating breakdown factor is shown in table below . They are evaluated to be 0.2mA/m² in accordance with ISO 155— (DEP 30.10.73.31 Gen.) Feb. 2013. Again, it should be noted that three relevant breakdowns used for design. These included,

- The Initial Breakdown
- The Mean Breakdown
- The Final Breakdown

The current need for group of pipelines which will be hooked at the manifold from same field or area is given by equation 2

$$I_{cmf} = \sum_1^n I_c \text{ —————eq. 2}$$

7.3.3 Sizing of Transformer – Rectifiers

7.3.4 Transformer –Rectifier for group of pipelines (n)

The transformer rectifier is used to convert AC to DC which is required for CP.

The size is based on the current demand and addition 50% and then using ohm's law, the voltage is obtained reference equations 3 & 4

$$I_{CTR} = \sum_1^n I_c + 50\% \sum_1^n I_c \text{ —————eq.3}$$

The Ohm' s law' which is given as

$$V=IR \text{ —————eq.4}$$

Where, V is Voltage, and R is Resistance. The resistance is based on the entire CP system resistance

Table 7.1: Surface Areas and Current Demand for the Pipelines CP.

SURFACE AREA & CURRENT DEMAND CALCULATIONS										
SN	P/LINE	LINE DIA.	LENGTH (m)	BARE SURFACE AREA (m ²)	MEAN EXPOSED SURFACE AREA (2%)	EXPOSED SURFACE AREA (LIFESPAN) (5%)	CORR. CURR. DENSITY (mA/m ² Bare Steel)	FINAL CURR. DEMAND (Amp)	MEAN CURR (Amp)	INITIAL CURR. (Amp)
1	NUN R – ELEP	24" (0.61m)	45,000	86,180.16	1,723.60	4,309.01	22	94.80	37.92	17.24
2	ELEP – AKAB	24" (0.61m)	28,500	59,368.56	1,187.37	2,968.43	22	65.31	26.12	11.87
3	OPUG –BBOU	24" (0.61m)	37,000	71,816.80	1,436.34	3,590.84	22	78.99	31.60	14.36
4	SEIB – NUNR	20" (0.51m)	20,000	31,918.58	638.37	3,191.86	22	70.22	14.04	6.4
5	BBOU – ELEP	16" (0.41m)	9,000	11,498.68	229.97	1,595.93	22	35.11	5.06	2.3
6	NUN R - MF	18" (0.46m)	10	47.87	0.9574	2.395	22	0.052	0.02	0.01
7	ELEP MF	12" (0.30m)	10	39.89	0.7978	1.995	22	0.044	0.02	0.01

8 Design For Temporary Protection System

The protection of the pipelines during the construction milestones is absolutely necessary. This will give progressive protection during the period of construction until the construction is completed and until the ICCP system is put in place.

Sacrificial Anode system is used for the temporary protection system.

The Anode choice is - High Output Magnesium (Mg) Anode

The Anode design life - 3 years

8.1 Design Data

8.1.1 Current Demand

The current demand is given in table 7.1 above on surface area and CP current demand calculations. The initial current demand will be utilised for further works as it will represent early stage of the current demand.

8.2 Characteristic of High Output Mg Anode (DEP 30.10.73.10 Gen)

Table 8.1: Characteristics of the High Output Magnesium Anode

SN	Description	Value
1	Design Life (years)	3
2	Anode Utilisation Factor (μ)	0.9
3	Anode Efficiency (ϵ)	0.5
4	Anode Protection Potential vs Cu/CuSO ₄ electrode (V)	0.950
5	Anode Closed Circuit Potential Seawater (V)	1.65
6	Anode Consumption Ca (Kg/Ay)	8
7	Length of design (m)	0.431
8	Diameter of Design Mg Anode D in m	0.1
9	Radius of Design r (meters)	0.05
10	Maximum Weight of Design Anode (KG)	6.1
11	Anode theoretical electrochemical capacity (Ah/kg)	2200

Table 8.2: High Output Magnesium Anode Composition

MAGNESIUM ANODE COMPOSITION		
S/N	ELEMENTS	PERCENTAGE
1	Cu	0.02% Maximum
2	Si	0.10% Maximum
3	Fe	0.003% Maximum
4	Ni	0.002% Maximum
5	Mn	1.0% Maximum
6	Mg	Remainder

Equation for calculation is as follows:

$$\text{Current (Amp)} = \text{Protected Area (m}^2\text{)} \times \text{Current density (mA/m}^2\text{)} / 1000 \quad \text{.....eqn 5}$$

$$\text{Weight (Kg)} = \text{Current (Amp)} \times \text{design life (years)} \times 8766 / \text{Capacity of material (Amph/Kg)} \\ \times 0.9 \quad \text{.....eqn 6}$$

$$\text{Number of Anodes} = \text{Weight Required} / \text{Individual Net Anode Weight} \quad \text{.....eqn 7}$$

$$\text{Number of Anodes} = \text{Current Required} / \text{Individual Current Output} \quad \text{.....eqn 8}$$

$$M_A = I_t \times t_d \times C_a / \mu \quad \text{.....eqn 9}$$

Where M_A = Mass of Anode (kg)

T_d = Design life (years)

I_t = Current demand (Ampere)

C_a = Consumption rate of Anode (kg/A – y)

μ = Utilization factor

Table 8.3: Number of Magnesium Anode Required Per Line for Temporary Protection (on the right column)

ANODE REQUIREMENTS							
S/N	LINE	DIAMETER		CURRENT DEMAND (It)	MASS OF ANODE FOR PROTECTION CURRENT (ma)	MASS OF SINGLE ANODE (ma)	No. OF ANODES
		Inch	mm				
1	SEIB- NUNR 20" x 20 Km	20	508	6.40	170.66	6.1	14
2	NUN R MF 18"x10 m	18	457.2	0.01	0.26	6.1	1
3	NUN R – ELEM 24"x 45 Km	24	609.6	17.24	459.7	6.1	38
4	OPUG – BBOU 24"x 37 Km	24	609.6	14.36	382.9	6.1	31
5	BBOU – ELEM 16"x 9 Km	16	406.4	2.30	61.33	6.1	5
6	ELEM MF 12"x 10 m	12	304.8	0.01	0.26	6.1	1
7	ELEM - AKAB 24"x 28.5 Km	24	609.6	11.87	284.88	6.1	26

NB : ELEM – AKAB is a coastal line and requires permanent Anodes likely Zinc bracelet type that will be factory manufactured and fitted.

The two Short MF lines may be protected with SA for Life Time.

8.3 Temporal Anode Spacing

Each line has its Anode requirement or number. These numbers are spaced equally along the line with 1st and last Anode being half the spacing of the end points of the line.

The Anode installation shall be in accordance to this document, specifications thereof and the manufacturer instructions. The Anodes shall be surrounded by a Bentonite / Gypsum backfill to avoid its polarisation.

NOTE : At completion before the commissioning of the project, the temporal sacrificial CP Mg Anode system shall be decommissioned and the impressed current CP system commissioned as may be the case.

8.4 Electrical Separation

The pipeline shall be electrically isolated from the piping as well as electrical and instrumentation earthing system. Isolation joints shall be installed,

1. At both extremities of the buried section.
2. At branch lines if they are not part of the main line.

The scope boundaries must be provided with electrical isolation to prevent interaction / electrical interference in the system. This will limit the possibility of electrical current licking in to adjacent installation.

9 Permanent CD Design

9.1 Design Basis

The Design shall be based on:

- Choice of CP system for each line (SA, ICCP, Hybrid System)
- All buried lines shall be impressed current system.
- All the coastal and submerged line shall be permanent SA (Bracelet anode type)
- Very short lines but buried shall be permanent SA Mg.

9.2 Detail Design Survey

The following detail survey should be under taken:

1. ROW – Chemical Analysis Survey
2. ROW – Resistivity Survey.
3. Ground bed Resistivity Survey for selection of ground bed locations either deep well or otherwise.

9.3 Choice of Electrical Generation System.

Other requirements are :

1. Cable marker
2. Cable Connections to the lines.
3. Removal of Coatings and Cleaning.
4. Thermal Weld Connection (CAD Welding).
5. Coating Repairs.
6. Holiday Test.
7. Junction Boxes
8. Test Posts / Stations

APPENDIX

- 1 Wellhead Gas Gathering Schematics & Flow Parameters to Gas Processing Plant
- 2 Proposed Pipeline Route
- 3 Pipeline Materials Selection

Appendix-1: Well Gas Gathering Schematics

Appendix 1	EA/SPDC-BFPCUGS-GEP/BDM-PFS-001-A01 Wellhead Gas Gathering Schematics & Flow Parameters to Gas Processing Plant	 EA SPDC-BFPC UGS-GEP BDM-PFS-01
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Appendix-2: Proposed Pipeline Route

Appendix 2	EA/SPDC-BFPCUGS-GEP/SUY-DWG-001-A01 Proposed Pipeline Route	 EA SPDC-BFPC UGS-GEP-SUY-DWG-0
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Appendix-3: Pipeline Materials Selection

Appendix 3	EA/SPDC-BFPC UGS-GEP/BDM-SCH-001-A01 Pipeline Materials Selection & Manifold Platforms Footprint Indication. BBOU & ELEP	 EA SPDC-BFPC UGS-GEP BDM-SCH-01
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